

MAT21-3

Multivariable Analysis

Albert School — 22 hours

Preface

In MAT11-1 you learned single-variable calculus: a derivative $f'(a)$ measures the slope of a curve, $f'(c) = 0$ locates its peaks and valleys, the second derivative tells a maximum from a minimum, and a Taylor expansion approximates a function near a point. Almost everything in this course is the faithful generalisation of those ideas to functions that depend on **several** inputs at once — the functions that describe real models, where a cost depends on many parameters rather than one.

Two things change when there is more than one variable. First, “the slope” is no longer a single number: at each point a function has a slope in every direction, and packaging all of them produces a **vector**, the gradient, rather than a scalar. Second, “the second derivative” becomes a **matrix**, the Hessian, and classifying a critical point becomes a question about that matrix — which is exactly where the determinants and eigenvalues from MAT21-1 (Linear Algebra II) re-enter. Keep both prerequisites in mind: the dot product, vectors, and matrices from linear algebra are the language; the limits, derivatives, and Taylor expansions from MAT11-1 are the content being lifted.

The payoff is the last third of the course. Once we can compute a gradient and read a Hessian, we can state and run **gradient descent**, the algorithm that trains essentially every modern machine-learning model, and we can say precisely when it converges and how fast. Multivariable analysis is the bridge from calculus to optimization and machine learning.

We work operationally throughout: domains are described by the inequalities that make a formula valid, with no formal topology. Most pictures live in $\mathbb{R}^2$ (two inputs, one output, so a surface over the plane), occasionally $\mathbb{R}^3$; the formulas are written so that the pattern to n variables is visible.

Contents

Preface	2
1 Functions of several variables	3
1.1 Graphs and level sets	3
2 Partial derivatives	4
2.1 Geometric meaning: slopes of cross-sections	5
3 Differentiability	5
4 The gradient	6
4.1 Directional derivatives	6
4.2 Steepest ascent	7
5 Chain rule and orthogonality to level curves	7
5.1 The chain rule along a curve	7
5.2 The gradient is orthogonal to level curves	7
5.3 Tangent line and tangent plane	8
6 The Hessian matrix	8
6.1 Second-order Taylor expansion	9
7 Classifying critical points	10
7.1 The Monge criterion	10
7.2 Eigenvalue interpretation	11
8 Gradient descent	11
8.1 Conditioning and the rate of convergence	12
9 Local versus global extrema	13
10 Putting it together: a machine-learning cost function	13

1 Functions of several variables

A company's monthly cost might depend on the price it sets **and** the amount it advertises; the temperature in a room depends on three space coordinates. Each is a rule that turns several numbers into one number. That is the object of this course.

Definition 1 A **function of several variables** assigns a real number to each point of a set $D \subseteq \mathbb{R}^n$. We write $f : D \rightarrow \mathbb{R}$ and call D the **domain**: the set of points at which the defining expression makes sense. We focus on $n = 2$ and $n = 3$ and treat domains operationally — by reading off the inequalities a formula requires — with no formal topology.

Worked example — Reading a domain off a formula.

- $f(x, y) = \sqrt{1 - x^2 - y^2}$ needs $1 - x^2 - y^2 \geq 0$, so $D = \{(x, y) : x^2 + y^2 \leq 1\}$, the closed unit disk.
- $g(x, y) = \ln(xy)$ needs $xy > 0$, so D is the union of the first and third quadrants (where x and y have the same sign).
- $h(x, y, z) = 1/(x^2 + y^2 + z^2)$ needs the denominator nonzero, so $D = \mathbb{R}^3 \setminus \{(0, 0, 0)\}$.

The recipe is exactly the single-variable one (no division by zero, no square root of a negative, no log of a non-positive), applied to the whole expression.

1.1 Graphs and level sets

For one variable the graph of f is a curve in the plane. For two variables it is a **surface** in space.

Definition 2 The **graph** of $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is the surface

$$\{(x, y, f(x, y)) : (x, y) \in D\} \subseteq \mathbb{R}^3,$$

i.e. the height $z = f(x, y)$ plotted above each point (x, y) of the domain.

Drawing a surface on paper is awkward, so we use the cartographer's trick: a contour map. Slice the surface at a fixed height c and project the slice down onto the domain.

Definition 3 A **level set** of f at value c is the set of domain points where f equals c :

$$\{(x, y) : f(x, y) = c\}.$$

In $\mathbb{R}^2$ this is a **level curve** (a contour line); in $\mathbb{R}^3$ a **level surface**.

Worked example — Two surfaces to keep in mind. These two examples recur throughout the course, so meet them now.

- The **paraboloid** $f(x, y) = x^2 + y^2$ is a bowl. Its level curves $x^2 + y^2 = c$ are concentric circles (radius $\sqrt{c}$, for $c \geq 0$) — tightly packed where the bowl is steep, far apart where it is shallow.
- The **saddle** $f(x, y) = x^2 - y^2$ rises along the x -axis and falls along the y -axis. Its level curves $x^2 - y^2 = c$ are hyperbolas, and the level set $c = 0$ is the pair of diagonal lines $y = \pm x$.

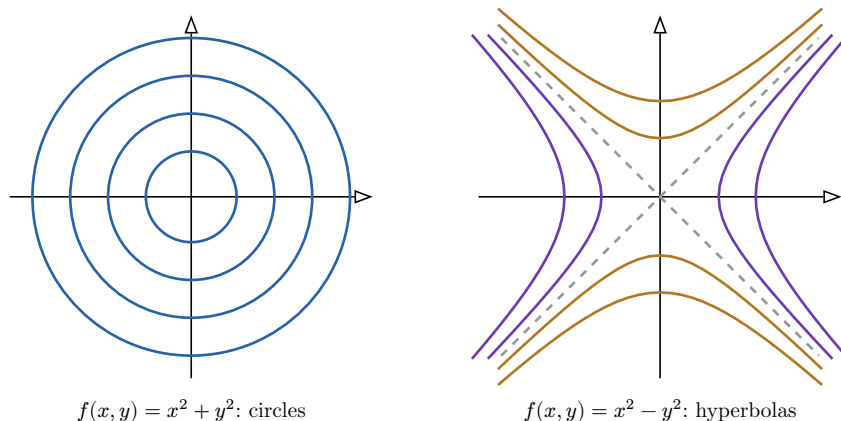


Figure 1: Contour maps. Left: the bowl $x^2 + y^2$ has circular level curves, closing in on the minimum at the centre. Right: the saddle $x^2 - y^2$ has hyperbolic level curves; the dashed diagonals are the level set $f = 0$ separating the rising region (purple, opening left/right) from the falling region (orange, opening up/down).

2 Partial derivatives

How does $f(x, y)$ change as we nudge the inputs? The simplest experiment is to freeze all variables but one and watch the rest — turning the multivariable function back into a single-variable one we already know how to differentiate.

Worked example — Freeze one variable. Take $f(x, y) = x^2y$. Hold y fixed at, say, $y = 3$: the function of x alone is $x \mapsto 3x^2$, whose derivative is $6x$. In general, treating y as a constant, the rate of change in the x -direction is $2xy$. Holding x fixed instead, f is linear in y with rate x^2 .

Definition 4 Let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ and $(a, b) \in D$. The **partial derivatives** of f at (a, b) are

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}, \quad \frac{\partial f}{\partial y}(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h},$$

when these limits exist. The definition extends to $\mathbb{R}^3$ and beyond, one variable at a time.

In practice you never compute the limit. To get $\partial f / \partial x$, differentiate the formula with respect to x using all the ordinary rules, treating every other variable as a constant.

Worked example — Computing partials. For $f(x, y) = x^2y + \sin(y)$,

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + \cos(y).$$

For $g(x, y) = e^{xy}$, the chain rule (with y constant) gives $\partial g / \partial x = ye^{xy}$ and, symmetrically, $\partial g / \partial y = xe^{xy}$.

Common pitfall The notation is ∂ (“partial”), not the straight d of ordinary derivatives, precisely to signal **which** variable is moving while the others stand still. Writing df/dx for a multivariable function hides that choice and is a frequent source of confusion later, in the chain rule.

2.1 Geometric meaning: slopes of cross-sections

Cutting the surface $z = f(x, y)$ with the vertical plane $y = b$ leaves a curve — the cross-section above the line $y = b$. The partial derivative $\partial f / \partial x(a, b)$ is exactly the slope of that curve at $x = a$. Likewise $\partial f / \partial y(a, b)$ is the slope of the cross-section cut by $x = a$. Each partial derivative is an ordinary single-variable slope, taken along one of the two coordinate directions.

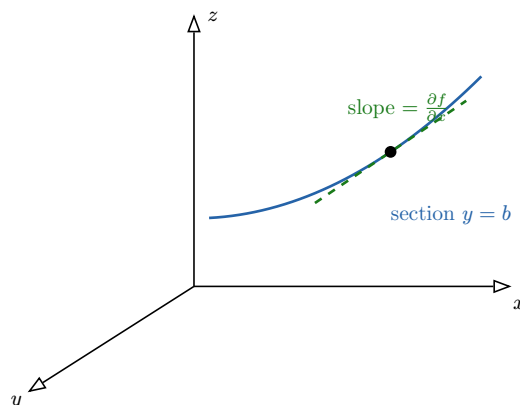


Figure 2: The cross-section of the surface above the line $y = b$ is a plane curve. Its slope at the point of interest is the partial derivative $\partial f / \partial x$. The other partial is the slope of the cross-section in the perpendicular (y) direction.

3 Differentiability

Having both partial derivatives means f behaves well along the two coordinate axes. Surprisingly, that is **not** enough to guarantee f behaves well in every direction — or even that it is continuous. **Differentiability** is the stronger property we actually want: that near (a, b) the surface is well approximated by a flat plane.

Definition 5 f is **differentiable** at (a, b) if there is a linear map L such that

$$f(a + h, b + k) = f(a, b) + L(h, k) + o(\|(h, k)\|) \quad \text{as } (h, k) \rightarrow (0, 0).$$

When it exists, L is forced to be $L(h, k) = \frac{\partial f}{\partial x}(a, b)h + \frac{\partial f}{\partial y}(a, b)k$, the **best linear approximation** to the change in f .

Checking that definition directly is painful. Fortunately there is a simple sufficient condition that covers every function built from the standard library.

Theorem 6 ($C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous) If the partial derivatives of f exist and are **continuous** on a neighbourhood of (a, b) — we say f is C^1 there — then f is differentiable at (a, b) . And if f is differentiable at (a, b) , then f is continuous at (a, b) .

So in practice: **if you can write down the partial derivatives and they are continuous formulas, f is differentiable.** The theorem matters mostly because its converses both fail, and the failures are instructive.

Common pitfall Existence of both partial derivatives does **not** imply differentiability, nor even continuity. For

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases},$$

both partials exist at the origin and equal 0 (along each axis f is identically 0). Yet along the line $y = x$ we get $f(x, x) = \frac{1}{2}$ for all $x \neq 0$, so f does not even tend to 0 at the origin: it is discontinuous there, hence not differentiable. The moral: partial derivatives probe only the two axis directions, and a function can hide bad behaviour along a diagonal.

Remark The chain of implications $C^1 \implies \text{differentiable} \implies \text{continuous}$ runs one way only. Each arrow has a counterexample for its converse: the function above is continuous-failing-from-partials, and one can also build functions that are differentiable but not C^1 . For this course, C^1 is the box you tick to use every tool that follows.

4 The gradient

The two partial derivatives are the rates of change along the axes. Collecting them into one vector gives a single object that controls the rate of change in **every** direction at once.

Definition 7 The **gradient** of a differentiable function f at a point is the vector of its partial derivatives,

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \quad (\text{and analogously in } \mathbb{R}^3 \setminus).$$

(The symbol ∇ is read “nabla” or “del”.)

Worked example — A gradient is a vector field. For $f(x, y) = x^2 + y^2$, $\nabla f = (2x, 2y)$. At the point $(1, 0)$ it is $(2, 0)$; at $(1, 1)$ it is $(2, 2)$. The gradient changes from point to point: it assigns a vector to each point of the domain.

4.1 Directional derivatives

The partials measure change along x and y . What about an arbitrary direction, say “north-east”?

Definition 8 Let u be a **unit** vector. The **directional derivative** of f at a point p in the direction u is

$$D_u f(p) = \lim_{h \rightarrow 0} \frac{f(p + hu) - f(p)}{h},$$

the rate of change of f as we move away from p along u .

Computing this from the limit would be tedious. The gradient does it for us with a single dot product.

Proposition 9 (Directional derivative via the gradient) If f is differentiable at p , then for every unit vector u ,

$$D_u f(p) = \nabla f(p) \cdot u.$$

Worked example. For $f(x, y) = x^2 + y^2$ at $p = (1, 1)$ we have $\nabla f(p) = (2, 2)$. The rate of increase heading north-east, $u = (1, 1)/\sqrt{2}$, is $(2, 2) \cdot (1, 1)/\sqrt{2} = 4/\sqrt{2} = 2\sqrt{2}$. Heading east, $u = (1, 0)$, the rate is just 2.

Common pitfall The formula $D_u f = \nabla f \cdot u$ assumes u is a **unit** vector. If you plug in a direction of length 2, you get twice the true rate. Always normalise the direction first: replace v by $v/\|v\|$.

4.2 Steepest ascent

Write the dot product with its angle: $D_u f(p) = \nabla f(p) \cdot u = \|\nabla f(p)\| \cos \theta$, where θ is the angle between u and the gradient. This is largest when $\cos \theta = 1$, i.e. when u points **along** the gradient.

The gradient points in the direction of steepest ascent, and its magnitude $\|\nabla f(p)\|$ is the maximal rate of increase. Pointing opposite to it, $-\nabla f(p)$, gives steepest **descent** — the idea the whole optimization half of the course rests on. Perpendicular to it ($\cos \theta = 0$), f does not change at all — which is the next section.

5 Chain rule and orthogonality to level curves

5.1 The chain rule along a curve

Suppose a point moves through the plane along a path $\gamma(t) = (x(t), y(t))$ and we watch the value $f(\gamma(t))$ it sees. How fast does that value change? The single-variable chain rule generalises into a dot product.

Theorem 10 (Chain rule along a curve) Let f be differentiable and let $\gamma(t) = (x(t), y(t))$ be a differentiable curve. Then $t \mapsto f(\gamma(t))$ is differentiable and

$$\frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t) = \frac{\partial f}{\partial x} x'(t) + \frac{\partial f}{\partial y} y'(t).$$

Worked example. Let $f(x, y) = x^2 y$ and $\gamma(t) = (\cos t, \sin t)$, the unit circle. Then $\nabla f = (2xy, x^2) = (2 \cos t \sin t, \cos^2 t)$ and $\gamma'(t) = (-\sin t, \cos t)$, so

$$\frac{d}{dt} f(\gamma(t)) = 2 \cos t \sin t \cdot (-\sin t) + \cos^2 t \cdot \cos t = \cos^3 t - 2 \cos t \sin^2 t.$$

5.2 The gradient is orthogonal to level curves

Now run a curve that stays **on a single level set** — so f never changes along it. The chain rule immediately forces the gradient perpendicular to the curve.

Proposition 11 (Gradient orthogonal to level curves) At any point p , the gradient $\nabla f(p)$ is orthogonal to the level curve of f through p (assuming $\nabla f(p) \neq 0$).

Proof. Let γ parametrize the level curve through $p = \gamma(t_0)$, so $f(\gamma(t)) = c$ is constant. Differentiating with the chain rule gives $0 = \nabla f(\gamma(t)) \cdot \gamma'(t)$ for all t . At t_0 the vector $\gamma'(t_0)$ is tangent to the level curve, and the equation says it is orthogonal to $\nabla f(p)$. $\square$

This is the geometric counterpart of steepest ascent: along a contour the value is flat (rate 0), so the uphill direction must be exactly perpendicular to the contour.

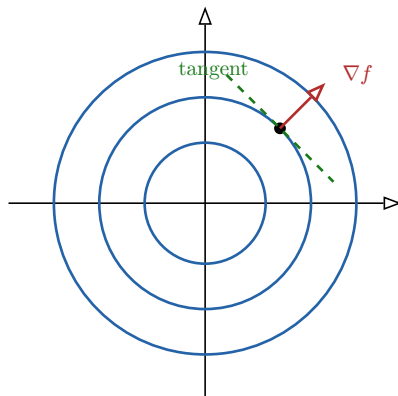


Figure 3: Level curves of $f(x, y) = x^2 + y^2$ (blue). At the marked point the gradient (red) points radially outward, straight uphill and exactly perpendicular to the contour; the tangent to the contour (green) is the one direction in which f does not change.

5.3 Tangent line and tangent plane

Orthogonality gives equations directly. The level curve $f(x, y) = c$ through (a, b) has, as tangent line, the line through (a, b) perpendicular to $\nabla f(a, b)$:

$$\frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b) = 0.$$

For the **graph** $z = f(x, y)$ of a differentiable function, the tangent plane at (a, b) is the first-order Taylor approximation — the flat sheet that best matches the surface there:

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

Worked example — Tangent plane to a paraboloid. For $f(x, y) = x^2 + y^2$ at $(1, 2)$: $f(1, 2) = 5$, $\nabla f = (2x, 2y)$ gives $(2, 4)$, so the tangent plane is $z = 5 + 2(x - 1) + 4(y - 2) = 2x + 4y - 5$.

Common pitfall Keep the two objects apart. The **tangent line** lives in the input plane and describes a level curve; its equation is set to 0. The **tangent plane** lives in space and describes the graph; its equation gives z . Both are built from the same partial derivatives, but they answer different questions.

6 The Hessian matrix

Single-variable calculus uses f'' to tell a maximum from a minimum. With several variables there are several second derivatives, and the natural place to store them is a matrix — bringing us back to the linear algebra of MAT21-1.

Definition 12 The **second-order partial derivatives** of f are the partials of its partials:

$$\frac{\partial^2 f}{\partial x^2}, \quad \frac{\partial^2 f}{\partial y^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right).$$

The **Hessian matrix** of f at a point collects them:

$$H_f = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}.$$

Worked example. For $f(x, y) = x^2y + y^3$: $\partial f/\partial x = 2xy$ and $\partial f/\partial y = x^2 + 3y^2$, so

$$\frac{\partial^2 f}{\partial x^2} = 2y, \quad \frac{\partial^2 f}{\partial y^2} = 6y, \quad \frac{\partial^2 f}{\partial x \partial y} = 2x = \frac{\partial^2 f}{\partial y \partial x}, \quad H_f = \begin{pmatrix} 2y & 2x \\ 2x & 6y \end{pmatrix}.$$

Notice the two mixed partials came out equal. That is no accident.

Theorem 13 (Schwarz) If the second-order partial derivatives of f are continuous near a point (i.e. f is C^2 there), then the mixed partials are equal:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

In particular the Hessian is a **symmetric** matrix.

Symmetry is what lets us use real eigenvalues later: a symmetric real matrix always has real eigenvalues and orthogonal eigenvectors.

Common pitfall Schwarz's theorem needs the second partials to be **continuous**. For exotic piecewise functions the mixed partials can genuinely differ at a point. Every function you meet in this course is C^2 on its domain, so the Hessian is symmetric — but the hypothesis is real, not decorative.

6.1 Second-order Taylor expansion

In one variable, $f(a+h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$. The multivariable version replaces f' by the gradient and f'' by the Hessian, with vectors and a matrix product.

Proposition 14 (Second-order Taylor expansion) For a C^2 function f , with displacement $h = (h_1, h_2)$,

$$f(p+h) = f(p) + \nabla f(p) \cdot h + \frac{1}{2} h^T H_f(p) h + o(\|h\|^2).$$

The quadratic term $\frac{1}{2}h^T H_f h$ is the multivariable “ $\frac{1}{2}f''(a)h^2$ ”. At a **critical point**, where $\nabla f(p) = 0$, the linear term vanishes and this quadratic form alone decides whether f curves up, down, or both — which is precisely the classification problem of the next chapter.

7 Classifying critical points

A peak, a valley, or a mountain pass: at all three the ground is momentarily level, so the gradient is zero. Telling them apart is the job of the Hessian.

Definition 15 A **critical point** (or stationary point) of f is a point where $\nabla f = 0$, i.e. a simultaneous solution of

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0.$$

Worked example — Finding critical points. For $f(x, y) = x^2 + y^2 - 4x$: setting $\nabla f = (2x - 4, 2y) = (0, 0)$ gives the single critical point $(2, 0)$. For the saddle $f(x, y) = x^2 - y^2$, $\nabla f = (2x, -2y) = 0$ gives the origin.

Around a critical point, the second-order Taylor expansion is governed by the quadratic form $h^T H_f h$. Three shapes are possible, illustrated by the three benchmark functions.

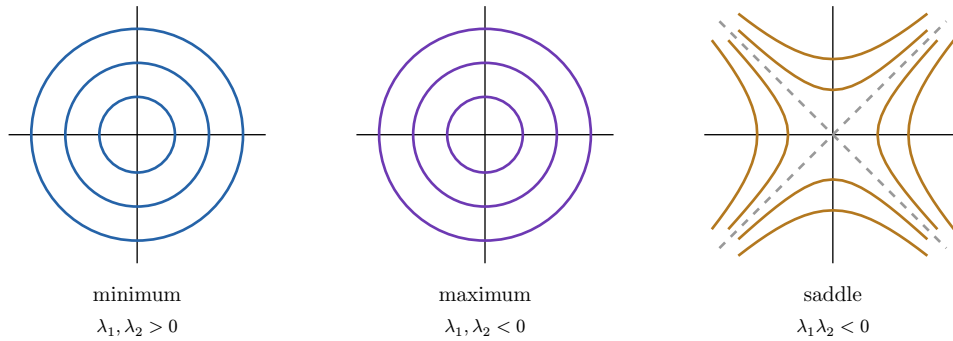


Figure 4: The three critical-point types, by their contours. Minimum and maximum both have nested closed level curves (the value rises out of, or falls away from, the centre); a saddle has hyperbolic contours that cross at the point. The sign pattern of the Hessian eigenvalues λ_1, λ_2 labels each case.

7.1 The Monge criterion

For two variables the classification reduces to three numbers read off the Hessian.

Definition 16 At a critical point write

$$r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}, \quad t = \frac{\partial^2 f}{\partial y^2}, \quad D = rt - s^2 = \det H_f.$$

Theorem 17 (Monge criterion) At a critical point, with $D = rt - s^2$:

- $D > 0$ and $r > 0$: **local minimum**;
- $D > 0$ and $r < 0$: **local maximum**;
- $D < 0$: **saddle point**;
- $D = 0$: the test is **inconclusive** (higher-order behaviour decides).

Worked example — Applying Monge. For $f(x, y) = x^2 + y^2 - 4x$ at the critical point $(2, 0)$: $r = 2, t = 2, s = 0$, so $D = 4 > 0$ and $r > 0$ — a local minimum. For the saddle $f(x, y) =$

$x^2 - y^2$ at the origin: $r = 2$, $t = -2$, $s = 0$, so $D = -4 < 0$ — a saddle point, as the name promised.

Common pitfall Two traps. First, the test is only valid **at a critical point**; computing D where $\nabla f \neq 0$ is meaningless. Second, when $D > 0$ you must still check the sign of r to separate a minimum from a maximum — and $D = 0$ tells you **nothing**, not “it is a saddle”. Reach for higher-order analysis there.

7.2 Eigenvalue interpretation

The Monge criterion is really a statement about the Hessian’s eigenvalues in disguise. Because H_f is symmetric, it has real eigenvalues λ_1, λ_2 , and the quadratic form $h^T H_f h$ stretches space by λ_i along the eigenvector directions. So:

- both eigenvalues positive (H_f positive definite): the form is positive in every direction $\rightarrow$ **local minimum**;
- both negative (H_f negative definite): $\rightarrow$ **local maximum**;
- eigenvalues of opposite sign (indefinite): up one way, down another $\rightarrow$ **saddle point**.

This recovers Monge exactly, because

$$D = \det H_f = \lambda_1 \lambda_2, \quad r + t = \text{tr } H_f = \lambda_1 + \lambda_2.$$

$D > 0$ means the eigenvalues share a sign (definite, an extremum), and then the trace — or equivalently r — picks which sign; $D < 0$ means opposite signs (indefinite, a saddle). The eigenvalue picture is the one that generalises to n variables, where “ D ” alone is no longer enough.

8 Gradient descent

We now have everything needed for the algorithm that makes the course matter. Steepest **descent** is the direction $-\nabla f$; to minimise f , repeatedly take a step that way.

Definition 18 Gradient descent minimises f by the iteration

$$x_{n+1} = x_n - \alpha \nabla f(x_n),$$

from an initial point x_0 , where $\alpha > 0$ is the **step size** (or **learning rate**).

Worked example — One run by hand. Minimise $f(x, y) = x^2 + y^2$ with $\alpha = 0.1$ from $x_0 = (1, 1)$. Here $\nabla f = (2x, 2y)$, so the update multiplies the point by $(1 - 2\alpha) = 0.8$:

$$x_1 = (0.8, 0.8), \quad x_2 = (0.64, 0.64), \quad \dots, \quad x_n = (0.8^n, 0.8^n) \rightarrow (0, 0).$$

The iteration marches into the minimum at the origin, the rate set by α .

The step size is the one knob, and it must be tuned.

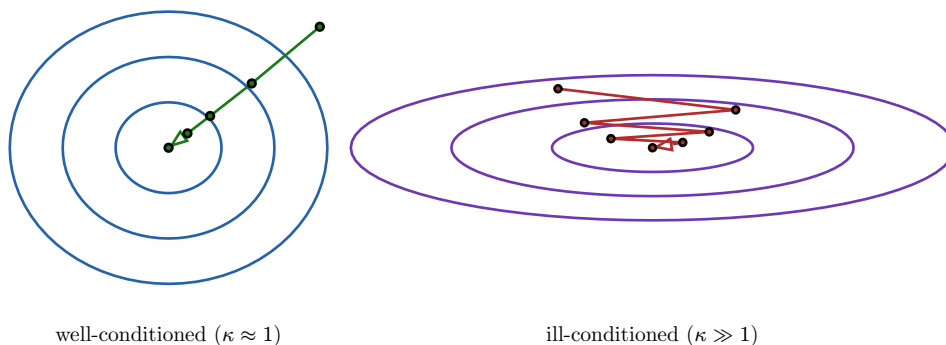


Figure 5: Gradient descent on quadratic bowls, drawn on their level curves. Left: near-circular contours, the steps head almost straight to the minimum. Right: elongated contours from an ill-conditioned Hessian force a slow zigzag — the step that suits the steep direction overshoots it while crawling along the shallow one.

Common pitfall The step size is a balancing act. **Too small** and the iteration creeps, needing many steps. **Too large** and a step overshoots the valley, so the iterates can oscillate or even diverge — in the example above, any $\alpha \geq 1$ makes $|1 - 2\alpha| \geq 1$ and the point fails to shrink toward the origin. There is no universally “right” α ; it depends on the function’s curvature.

Proposition 19 (Fixed points are critical points) If gradient descent converges, $x_n \rightarrow x^*$, and ∇f is continuous, then taking the limit in $x_{n+1} = x_n - \alpha \nabla f(x_n)$ gives $x^* = x^* - \alpha \nabla f(x^*)$, hence $\nabla f(x^*) = 0$: the limit is a critical point of f .

8.1 Conditioning and the rate of convergence

The zigzag above is not bad luck; it is dictated by the Hessian eigenvalues. Near a minimum, f looks like a quadratic bowl whose cross-section steepness in each eigen-direction is the eigenvalue λ_i .

Definition 20 For a quadratic with positive Hessian eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$, the **condition number** is

$$\kappa = \frac{\lambda_1}{\lambda_n},$$

the ratio of the steepest to the shallowest curvature.

When $\kappa \approx 1$ the bowl is nearly round, every direction curves alike, and one well-chosen α suits them all: convergence is fast. When κ is large — an **ill-conditioned** Hessian — the level sets are long thin ellipses; a step small enough not to overshoot the steep direction barely moves along the shallow one, so progress is slow. Concretely the error contracts by roughly a factor $\frac{\kappa-1}{\kappa+1}$ per step, which is near 1 (slow) exactly when κ is large.

Why this matters for machine learning. Training minimises a cost over many parameters; its Hessian is typically ill-conditioned, so plain gradient descent crawls. This single fact

motivates feature scaling, better step-size schedules, momentum, and adaptive methods — all attempts to fight a large κ . The eigenvalue analysis here is the diagnosis those methods treat.

9 Local versus global extrema

The Monge and eigenvalue tests are **local**: they examine f in a small neighbourhood and so can only certify a **local** extremum. A local minimum is lowest only nearby; a **global** minimum is lowest over the whole domain. A function can have many local minima at different heights — and gradient descent, following the slope downhill, may halt at whichever one it happens to fall into.

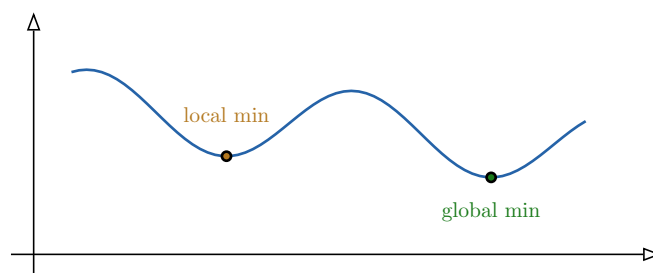


Figure 6: A function with two minima at different depths (a 1-D slice). Both are local minima — the slope is zero and the curve bends up — but only the deeper one is global. A local test cannot tell them apart; descent from the left would settle in the shallow well.

The cure is a **global** hypothesis on f : convexity, lifted from MAT11-1’s “lies above its tangent lines” to several variables.

Definition 21 f is **convex** on a convex domain if for all points p, q and all $t \in [0, 1]$,

$$f(tp + (1 - t)q) \leq tf(p) + (1 - t)f(q)$$

(the graph lies below every chord). For a C^2 function, this is equivalent to the Hessian being **positive semidefinite** everywhere (all eigenvalues ≥ 0).

Proposition 22 (Convexity makes local global) If f is convex, then every local minimum is a **global** minimum. In particular any critical point of a convex function is a global minimum.

Convexity is the property that makes gradient descent trustworthy: there are no spurious local minima to get stuck in, so wherever the algorithm settles is the best answer. Designing a model so its cost function is convex (or nearly so) is therefore a central modelling concern.

10 Putting it together: a machine-learning cost function

Every tool in this course converges on one workflow. Given a cost function f of some parameters, do the following.

1. **Gradient.** Compute ∇f and solve $\nabla f = 0$ for the critical point(s).

2. **Classify.** At each critical point form the Hessian; use the Monge criterion ($D = rt - s^2$) or its eigenvalues to label it a minimum, maximum, or saddle.
3. **Descend.** Run $x_{n+1} = x_n - \alpha \nabla f(x_n)$; use the Hessian eigenvalues (the condition number κ) to anticipate whether convergence is fast or slow, and tune α accordingly.
4. **Global?** If f is convex (Hessian positive semidefinite everywhere), the minimum found is the global one and descent reaches the optimal parameters.

Worked example — A least-squares cost. A one-parameter pair model with cost $f(a, b) = (a + b - 1)^2 + (a - b - 2)^2$ has $\nabla f = (2(a + b - 1) + 2(a - b - 2), 2(a + b - 1) - 2(a - b - 2))$. Setting it to zero gives $a + b = 1$ and $a - b = 2$, so the unique critical point is $(a, b) = (\frac{3}{2}, -\frac{1}{2})$. The Hessian is the constant $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ — eigenvalues 4, 4 > 0, condition number $\kappa = 1$. So the cost is convex, the critical point is the **global** minimum, and gradient descent converges quickly from any start. This is the shape of every ordinary least-squares problem.

11 General knowledge: vector-valued maps and the Jacobian

So far the output of f was a single number. When the output is itself a vector — m numbers depending on n inputs — the gradient generalises to a matrix. This section is **general knowledge**: you should know the definitions and the chain-rule statement, as the downstream Deep Learning and Optimization courses assume them.

Definition 23 Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $F = (F_1, \dots, F_m)$. The **Jacobian matrix** of F at a point is the $m \times n$ matrix of all first-order partials,

$$J_F = \left(\frac{\partial F_i}{\partial x_j} \right)_{1 \leq i \leq m, 1 \leq j \leq n} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}.$$

Its i -th row is the gradient of the component F_i . For a scalar function ($m = 1$) the Jacobian is the single row ∇f .

Worked example. For $F(x, y) = (x^2y, x + \sin y)$ from $\mathbb{R}^2$ to $\mathbb{R}^2$,

$$J_F = \begin{pmatrix} 2xy & x^2 \\ 1 & \cos y \end{pmatrix}.$$

Theorem 24 (Multivariable chain rule) For composable differentiable maps F and G , the Jacobian of the composition is the **product of the Jacobians**:

$$J_{G \circ F}(p) = J_G(F(p)) J_F(p).$$

This is the single-variable $(g \circ f)' = g'(f(x))f'(x)$ with numbers replaced by matrices and multiplication by matrix multiplication — and the matrix product is read right-to-left, innermost map first.

This statement is exactly the engine of **backpropagation**: a deep network is a long composition of layers, and its overall Jacobian is the product of the layers' Jacobians, computed efficiently from output back to input. Recognising the chain rule as “multiply the Jacobians” is the prerequisite the Deep Learning Mathematics and Optimization courses build directly upon.